

Module Review: Nitrogen-Regulatory Phosphotransferase System (PTS^{Ntr}) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Local bucket:** kegg:ppu02060 — KEGG "Phosphotransferase system (PTS)" **Module area:** transport_motility_signaling **Date:** 2026-09-01

1. Executive summary

The nitrogen-regulatory PTS (PTS^{Ntr}) module is **fully satisfiable** in *P. putida* KT2440, and every step is supported by **direct experiments in the target strain** (not homology transfer). The three module steps map cleanly to three genes:

Module step	Protein	Gene (symbol)	Locus	UniProt	KEGG KO / EC
1. PEP-dependent entry	Enzyme I ^{Ntr} (EI ^{Ntr})	<i>ptsP</i>	PP_5145	Q88CN5	K08484 / EC 2.7.3.9
2. Phosphocarrier relay	NPr	<i>ptsO</i> (KEGG symbol legacy " <i>ptsH</i> ")	PP_0948	Q88PA2	K08485 (NPr)
3. Terminal regulator	EIIA ^{Ntr}	<i>ptsN</i>	PP_0950	Q88PA0	K02806

The primary phosphate flow **PEP → EI^{Ntr} → NPr → EIIA^{Ntr}** was demonstrated in KT2440 by in-vivo phosphorylation-state assays ([P 18296519](#)), and each protein has genetically confirmed function in the strain (PMIDs 17416664, 11489866, 21236318, 26224366, 22434849).

Three curation actions dominate this review: 1. The KEGG bucket **ppu02060 conflates two biologically distinct processes** — the fructose/sugar PTS (*fruBKA*) and the regulatory PTS^{Ntr}. Only the latter belongs to this module; **fruB/fruK/fruA should be excluded** (*fruK* is a glycolytic 1-phosphofructokinase, not a PTS component at all). 2. **PP_0948 carries a misleading legacy symbol "*ptsH*"/"HPr"** but is the nitrogen-PTS **NPr** (KEGG ortholog K08485), confirmed by *rpoN*-operon synteny and His15 mutagenesis. Its symbol should be corrected to *ptsO/npr*. 3. Phosphate entry is **not strictly *ptsP*-exclusive** (a second EI-type enzyme "*mtp*" and FruB cross-talk provide redundancy), a caveat for the step-1 molecular-player annotation.

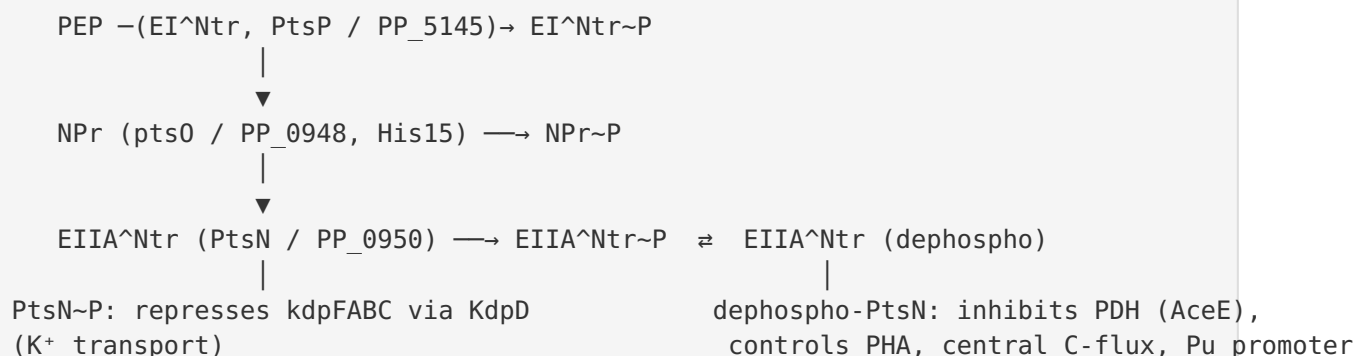
2. Target-organism pathway definition

What the module IS (in KT2440): a cytoplasmic, membrane-independent PEP-driven phosphorelay that functions **purely in signaling/regulation** — it does not transport or phosphorylate any sugar. PEP donates a phosphoryl group to EI^{Ntr} (PtsP), which relays via the small phosphocarrier NPr (His15) to the terminal regulator EIIA^{Ntr} (PtsN). The **phosphorylation state of EIIA^{Ntr}** (set by the cell's PEP/pyruvate ratio and C/N status) is read out by direct protein–protein interactions with downstream targets. This is the "N/C-sensing" or "nitrogen-related/metabolic" PTS.

Alternate names / database definitions: - PTS^{Ntr}, PTS(Ntr), "nitrogen-related PTS", "nitrogen-metabolic PTS", "N/C-sensing PTS", "regulatory PTS", "paralogous/branch PTS". - KEGG places all of these genes on map **ppu02060 "Phosphotransferase system (PTS)"**, which is an **overview transport map that also contains all sugar PTS proteins** — the map name is broader than this module.

Neighboring processes to keep SEPARATE from this module: - **Fructose PTS / PTS^{Fru}** (*fruBKA*, PP_0793–0795): carbohydrate uptake + fructose-1-P phosphorylation. Shares map ppu02060 and cross-talks with PTS^{Ntr} but is a distinct pathway. - **Glycolysis / fructose metabolism** (*fruK* = 1-phosphofructokinase). - **σ⁵⁴ (RpoN) regulon / nitrogen assimilation** (Ntr/GlnK–NtrBC systems): functionally connected but a separate module. Note NPr/PtsN are *genomically* embedded in the *rpoN* operon here, which can invite conflation. - **KdpD/KdpE two-component system and KdpFABC K⁺ transport:** a *target* of PtsN, not part of the phosphorelay module.

3. Expected step model



All three steps and both inter-step relationships in the provisional module outline are **confirmed in KT2440**.

4. Candidate genes and evidence

4.1 Genes that ARE the module (high confidence)

EI^Ntr — *ptsP* / **PP_5145** (Q88CN5; K08484; EC 2.7.3.9; 759 aa) - Role: PEP-dependent autophosphorylating Enzyme I of the Ntr branch; entry point of the relay. - Domain architecture: N-terminal **GAF** sensory domain + PEP-utilizer (PEP-utilizers_N/C) catalytic domains — the GAF domain is the presumptive metabolic-signal input (analogous to other EI^Ntr proteins). - Evidence: direct KT2440 genetics — part of the PEP→EI^Ntr→NPr→EIIA^Ntr flow

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ptsP/ptsO/ptsN act in concert to control PHA

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- Caveat: **phosphodonor redundancy** — a second EI-type ORF "*mtp*" was noted, and neither *ptsP* nor *mtp* single mutants abolished Pu carbon regulation

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FruB cross-talk can also phosphorylate downstream EIIA^{Ntr} in a *ptsP* background
([P 18296519](#)).

→ mark step 1 "covered (PtsP primary) with documented redundancy."

NPr — PP_0948 (Q88PA2; **KO K08485 = NPr**; 90 aa; PTS-HPr fold; active-site His15) - Role: phosphocarrier relay between EI^{Ntr} and EIIA^{Ntr}. - Evidence: H15A substitution abolishes ptsO function in carbon regulation of Pu

([P 11489866](#))

— direct proof of the phospho-His relay; coexpressed with *ptsN*, mapped adjacent to *rpoN*

([P 11489866](#));

part of the concerted PHA-controlling cascade

([P 17416664](#)).

- **Curation caveat (important):** the KEGG/GenBank symbol is "*ptsH*" and name "**Phosphocarrier protein HPr**", but the ortholog assignment (K08485) and all functional/synteny evidence identify it as the **NPr (*ptsO/npr*)**. KT2440 has **no separate canonical carbohydrate HPr** — the fructose branch's HPr activity is fused into the FruB multidomain protein

([P 18296519](#)).

→ **Recommend symbol correction PP_0948: ptsH → ptsO/npr.**

EIIA^{Ntr} — ptsN / PP_0950 (Q88PA0; K02806; 154 aa; *PTS_EIIA_2*) - Role: **terminal phosphorylation-state-dependent regulator acting by direct protein–protein interaction (no sugar phosphorylation)**. - **Direct KT2440 targets:** - *PtsN~P* represses *kdpFABC* (K⁺ transport) via direct interaction with the sensor kinase **KdpD**; K⁺-dependent

([P 26224366](#));

modeled in

([P 26159078](#)).

- **Dephospho-PtsN inhibits pyruvate dehydrogenase** (AceE, E1 subunit) — co-IP/MS, co-localization, activity assays

([P 21236318](#)).

- Controls **central-carbon flux** (elevated pyruvate-shunt flux in *ptsN* mutants;

(**P** 22434849)
 and **PHA** **accumulation**
 (**P** 17416664).
 - Modulates the **Pu/TOL biodegradation promoter** in response to carbon
 (**P** 11489866).
 (**P** 30209872)
 - Opposite phospho-forms act on different targets — a bona fide phosphorylation-state switch.

4.2 Candidate genes that are NOT this module (KEGG map overlap only)

fruB / **PP_0793** (Q88PQ5; multidomain EI-HPr-EIIA^{Fru}; EC 2.7.3.9) — fructose PTS; relevant only as a documented **cross-talk phosphodonor** to EIIA^{Ntr}

(**P** 18296519)
 (**P** 22708906)

Not a PTS^{Ntr} component. **fruK** / **PP_0794** (Q88PQ4; 1-phosphofructokinase) — a **glycolytic enzyme, not a PTS protein at all**; broad "Phosphofructokinase" annotation. Exclude. **fruA** / **PP_0795** (Q88PQ3; IIBC^{Fru} permease; EC 2.7.1.202) — fructose PTS permease. Not a PTS^{Ntr} component.

5. Gaps, ambiguities, and likely over-annotations

- **Module-boundary error in the source bucket:** ppu02060 mixes PTS^{Ntr} with the fructose PTS. The generic module boundary is correct in concept (it explicitly excludes sugar transport), but the *local candidate list* is contaminated with *fruBKA*. → `module_needs_revision` at the gene-membership level.
- **PP_0948 symbol over-propagation:** "*ptsH*/HPr" is a legacy mislabel; the protein is NPr. High-impact for curation because a curator could wrongly assign it to carbohydrate PTS or expect a separate NPr elsewhere (there is none). **Genome-wide KO check confirms this:** KT2440 has **zero** genes assigned to the canonical HPr ortholog **K02784** and **zero** to the

carbohydrate Enzyme I ortholog **K08483** — so the "HPr" label has no legitimate genomic referent, and FruB is the sole sugar-PTS Enzyme I.

- **No intra-genome paralog ambiguity:** each PTS^{Ntr} step is single-copy in KT2440 — *ptsP*=PP_5145 (K08484), *NPr*=PP_0948 (K08485), *ptsN*=PP_0950 (K02806). This contrasts with *Rhizobium leguminosarum* (two *ptsN* copies;

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so KT2440 module steps map unambiguously to one gene each.

- **"mtp" is not a distinct Enzyme I:** the only other PEP-utilizing enzyme in KT2440 is PEP synthase (*ppsA*, PP_2082, K01007); there is no second EI/EI^{Ntr} ortholog. Step-1 redundancy therefore most plausibly operates through FruB cross-talk rather than a bona fide second Enzyme I.
- **fruK broad EC/name:** "Phosphofructokinase" — ensure it is annotated as 1-phosphofructokinase (fructose-1-P → fructose-1,6-bisP) and not counted toward any PTS step.
- **EI^{Ntr} phosphodonor redundancy:** step-1 input is not strictly PtsP-exclusive — FruB cross-talk feeds the relay in a *ptsP* background

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and the historic "mtp" ORF

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has no distinct modern EI ortholog (see above). Curate step 1 as "PtsP primary, with documented redundancy."

- **PP_0949 (UPF0042 ATP-binding, RapZ/YhbJ homolog):** present in the operon; associated in other taxa with GlmZ sRNA / RNase adaptor regulation — outside this three-step module but a candidate for a separate module/review.
- **NPr's own direct targets:** in *E. coli*, dephospho-NPr regulates LpxD/LPS and envelope stress

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Whether NPr has PtsN-independent targets in KT2440 is **untested (uncertain species transfer)** — an open question.

- **Broader-organism inferences (flag as transfer, not direct):** *E. coli* cross-talk→Kdp

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Rhizobium PTS^{Ntr}→ABC transporters

(P 22340847);

P. aeruginosa PtsN redundant phosphorylation and virulence links

(P 37074168).

These are **weak-to-moderate** transfer to KT2440 and should not be asserted as KT2440 facts.

6. Module and GO-curation recommendations

Step status: | Module step | Gene | Status | |---|---|---| | 1. EI^{Ntr} PEP entry | *ptsP* / PP_5145 | **covered** (primary donor; note redundancy) | | 2. NPr relay | PP_0948 (*ptsO/npr*) | **covered** (correct the "*ptsH*" symbol) | | 3. EIIA^{Ntr} terminal regulator | *ptsN* / PP_0950 | **covered** | | Sugar PTS genes in bucket | *fruB/fruK/fruA* | **not_expected_in_module** (belong to PTS^{Fru} / glycolysis) |

Module document: Retain a dedicated PTS^{Ntr} module distinct from the KEGG ppu02060 overview map; explicitly list membership = {PP_5145, PP_0948, PP_0950} and explicitly exclude {PP_0793, PP_0794, PP_0795}. Record FruB and "mtp" as *cross-talk/redundant inputs*, not members.

GO-curation suggestions (KT2440-supported): - PP_5145 (PtsP): GO:0008965 phosphoenolpyruvate-protein phosphotransferase activity; GO:0009401 PTS; phosphorelay signal transduction. - PP_0948 (NPr): protein-histidine phosphotransfer / phosphorelay carrier; **request symbol change**; ensure it is NOT annotated as carbohydrate HPr. - PP_0950 (PtsN): protein-kinase-regulator/enzyme-regulator activity via KdpD interaction → **GO:0043266 regulation of potassium ion transport** (and negative regulation of K⁺ import); **negative regulation of pyruvate dehydrogenase (acetyl-transferring) activity / pyruvate metabolic process**; regulation of PHA/carbon storage. These are direct-experiment-supported and appropriate as IMP/IPI (KT2440). - Consider a **new/checked GO term** for "regulation of potassium ion transmembrane transport by protein-protein interaction with a sensor histidine kinase" if existing terms are too generic.

7. Genes to promote to full fetch-gene review

1. **PP_0948** (top priority) — resolve the *ptsH*→*ptsO*/*npr* symbol/name discrepancy; confirm NPr status and His15 annotation. (*direct evidence*:

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ortholog *K08485*)

2. **PP_5145 (*ptsP*)** — annotate GAF sensory domain, EC 2.7.3.9, and the redundancy caveat.
3. **PP_0950 (*ptsN*)** — attach the KdpD / PDH / PHA / central-C regulatory GO annotations.
4. **PP_0949** (UPF0042 / RapZ-YhbJ homolog) — clarify whether it belongs to a separate regulatory module.
5. "**mtp**" — flagged in

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as a second EI-type ORF, but a genome-wide KO check finds no distinct Enzyme I/EI^{Ntr} ortholog (only *ppsA*/PP_2082 PEP synthase); low priority — likely superseded by the FruB cross-talk explanation. Track only if resolving historical nomenclature.

8. Key references

- Pflüger & de Lorenzo, 2008. In-vivo cross-talk between nitrogen- and fructose-related PTS branches of *P. putida*. *J Bacteriol.*

P 18296519

(Direct: PEP→EI^{Ntr}→NPr→EI^{IA}^{Ntr} flow; only 5 PTS proteins; FruB cross-talk.)

- Cases, Velázquez & de Lorenzo, 2001. Role of *ptsO* in carbon inhibition of the Pu promoter. *J Bacteriol.*

P 11489866

(Direct: NPr His15 required; *ptsO*–*ptsN*–*rpoN* synteny; *ptsP*/*mtp* redundancy.)

- Velázquez et al., 2007. *PtsP*/*PtsO*/*PtsN* control PHA accumulation in *P. putida*.

P 17416664

(Direct: three-protein cascade intact.)

- Pflüger-Grau, Chavarría & de Lorenzo, 2011. EIIA^{Ntr} interplay with pyruvate dehydrogenase. *Mol Microbiol*.

[P 21236318](#)

(Direct: dephospho-PtsN inhibits PDH/AceE.)

- Chavarría et al., 2012. Regulatory tasks of the PTS in central carbon metabolism of *P. putida*.

[P 22434849](#)

(Direct: PtsN controls pyruvate-shunt flux.)

- Deuschle et al., 2015. PtsN–KdpD interplay controlling K⁺ transport.

[P 26224366](#)

(Direct: PtsN~P represses kdpFABC via KdpD.)

- Chavarría et al., 2013. Cra regulates cross-talk between the two PTS branches.

[P 22708906](#)

(Direct: FruB→EIIA^{Ntr} phosphotransfer.)

- Pérez-Pantoja et al., 2018. EIIA interplay with Pu promoter regulation.

[P 30209872](#)

- Wolf, Pflüger-Grau & Kremling, 2015. Modeling PtsN–KdpFABC interplay.

[P 26159078](#)

- *Transfer/context (non-KT2440)*: Lüttmann et al., 2015 (*E. coli* cross-talk→Kdp,

[P 26159077](#)

Lee et al., 2015 (*E. coli* NPr→LpxD/envelope stress,

[P 25701731](#)

Prell et al., 2012 (*Rhizobium* PTS^{Ntr}→ABC transporters,

[P 22340847](#)

Underhill et al., 2023 (*P. aeruginosa* PtsN redundancy/virulence,

[P 37074168](#)

Prepared for manual module satisfiability and gene-annotation curation. Direct KT2440 evidence is explicitly distinguished from cross-species inference throughout.

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